Highlights

Explainable production-adjusted electrical-energy baselines for industrial feed pelleting

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- Production-adjusted electricity baseline from 2,745 real feed-pelleting records
- Frozen Ridge model cuts test MAE by 14.42% relative to line-specific SEC
- TFT, N-HiTS, TimesFM 2.5 and Chronos-2 add no accuracy on identical plant records
- Exact additive decomposition traces each kWh estimate to recorded production inputs
- Fixed 90%/95% residual intervals cover 85.96% and 93.23% of later test records

Explainable production-adjusted electrical-energy baselines for industrial feed pelleting

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Abstract

A plant running an energy management system needs to know how much electricity an operation should have used given what it processed. In feed pelleting the usual answer, one specific energy consumption (SEC) per line, cannot tell an inefficient operation from an unusual product mix. We built a production-adjusted electricity baseline on 2,745 real records from three industrial lines, with dosed mass, dosing-batch count, product characteristics, line identity and calendar variables as inputs, and asked whether histories of completed operations, temporal models or pretrained foundation models add anything to it. Seven estimators were compared on 800 chronologically ordered development targets; the selected Ridge regression was then frozen, with its interval calibration, and evaluated once against line-specific SEC on 798 later records. Eight current-input, history-enabled, temporal and pretrained configurations were compared on 584 matched targets sharing 16 completed same-line operations as context. Ridge reduced test MAE from 61.29 to 52.45 kWh, 14.42% below SEC, on all three lines; this is a lower prediction error, not an

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energy saving. The Temporal Fusion Transformer reached 53.65 kWh against 53.55 kWh for current-input Ridge, with no evidence of paired improvement; N-HiTS, TimesFM 2.5 and Chronos-2 were less accurate. Every estimate decomposes exactly into its inputs, and fixed 90% and 95% residual intervals covered 85.96% and 93.23% of the test records, with one line systematically underestimated. The baseline is ready for retrospective assessment and for a digital twin; prospective use needs verified inputs, line-specific acceptance criteria and a new consecutive period.

Keywords: feed pelleting, industrial energy baseline, time-series foundation models, chronological validation, prediction intervals, explainable machine learning

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Nomenclature

Symbols

$y_i,\ E_i$	pelleting electrical energy of record i (kWh)
$\widehat{y}_i$	point estimate of y_i (kWh)
m_i	dosed mass of record i (t)
b_i	number of dosing batches of record i
$\ell_i,\ c_i,\ p_i,\ s_i$	pelleting line, product subfamily, presentation and bag/bulk category of record i
$h_i,\ d_i$	origin hour (h) and weekday (0–6) of record i
t_i	analytical origin of record i: earliest recorded stage start
a_j	recorded pelleting completion of record j
Δ_i	recorded pelleting duration of record i (h)
P_i^{avg}	duration-based average-power proxy of record i (kW)
z_i	standardized and one-hot encoded input vector of record i
$\widehat{b},\ \widehat{\beta}$	fitted intercept and coefficient vector of the Ridge model
$\widehat{s}_\ell$	training-period specific energy consumption of line ℓ (kWh/t)
$\mathcal{T},\ \mathcal{P}$	training set and partition interval
$\mathcal{H}_i^{(16)},\ \mathcal{H}_i$	16-record operating history and recent-residual history of target i
C_{ig}	contribution of variable group g to the estimate of record i (kWh)
δ_i	shrunk recent-residual correction (kWh)
q_α	calibrated half-width at miscoverage level α (kWh)
$L_i,\ U_i$	lower and upper interval limits of record i (kWh)
$r_i^{\text{excess}},\ B_i$	signed excess over the estimate (kWh) and out-of-interval flag

Abbreviations

EnB, EnPI	energy baseline, energy performance indicator
IPMVP	International Performance Measurement and Verification Protocol
MAE, RMSE, WAPE	mean absolute error, root mean squared error, weighted absolute percentage error
IS	interval score
MLP	multilayer perceptron
SEC	specific energy consumption (kWh/t)
TFT	Temporal Fusion Transformer
N-HiTS	Neural Hierarchical Interpolation for Time Series
XReg	covariate-regression adapter of TimesFM
H1, H2, H3	post-hoc extensions: line-dependent mass effect, product-dependent mass effect, recent-residual correction
GPU, CPU	graphics and central processing unit

1. Introduction

Two pelleting operations run on the same day can use very different amounts of electricity for entirely ordinary reasons: one processes more material, runs a different product, or takes place on another line. Before a plant can say anything about efficiency, those differences have to be accounted for. In feed manufacturing the usual tool is a specific energy consumption (SEC) in kWh/t. It is simple and easy to communicate, but it cannot separate the effects of product mix, line assignment, processed mass and batch structure, so an operation that looks wasteful may simply have been an unusual one. Energy management standards are explicit about the alternative: ISO 50001 asks for energy performance to be judged against a baseline normalized for the relevant variables, and ISO 50006 and the IPMVP describe how such baselines and performance indicators are built and used [1, 2, 3]. The stakes are not small. Food processing in the United States consumes about 75 TWh of electricity a year, and electricity accounts for roughly 30% of the reported energy use of UK food and beverage manufacturing [4, 5, 6].

Pelleting is a good place to build such a baseline because its electricity demand depends so strongly on what is being processed. Pellet quality, conditioning, feed composition, steam application, temperature, particle characteristics and operating conditions all shape pelleting behaviour and mechanical energy demand [7, 8, 9, 10, 11]. Those studies, however, are about material properties, process behaviour or pellet quality. None of them builds an operation-level electricity baseline from real plant records and tests it, in time order, against the reference a plant actually uses. That is the gap addressed here.

1.1. Related work

Regression-based baselines for measurement and verification go back at least to the monthly-data models of Kissock and Eger [12]. Automated baselines on interval data later showed that predictive accuracy and honest uncertainty decide whether a saving can be claimed at all [13, 14], and key-performance-indicator frameworks carry the same idea into production facilities [15]. In the feed industry itself, Sánchez et al. [16] built motor-level efficiency assessments, energy baselines and performance indicators for a balanced-feed plant (about 10 kWh/t at 500 t/day, with an estimated saving potential of 364,000 kWh/year), but at an aggregated process level rather than per operation. Wigger et al. [17] showed how AI models can be integrated into the energy management system of a manufacturing company from its existing data, with consumption forecasting and anomaly detection as the use cases; their object is the company's consumption series, not the individual operation and what it processed.

On the modelling side, regularized regression gives stable and interpretable estimates when predictors are correlated [18], while tree ensembles and neural networks capture nonlinearities and interactions [19]. The study closest to ours is that of Ekwaro-Osire et al. [20], who compared linear regression, tree-based methods and a multilayer perceptron on about 1,100 production batches with 67 input characteristics from an industrial compound-feed pelleting process and found that the more complex models did not necessarily reduce error. Because it relied on shuffled cross-validation, it could not say how the models behave on a later period, and it did not consider operating histories, pretrained temporal models, predictive uncertainty or record-level traceability. Panjapornpon et al. [21] predicted the energy efficiency of a petrochemical process with an explainable transfer-learning LSTM made robust to sensor faults, but on regularly sampled signals rather than on heterogeneous completed operations. In this journal, production energy under complex working conditions has been predicted with an attention-based recurrent network [22], the consumption of hot-die-forging lines with a hybrid deep architecture [23], and energy-prediction models have been deployed in manufacturing through asset administration shells [24]. Our setting differs in two respects: the inputs are the consolidated production descriptors that a plant database already holds rather than operating signals, and the question is whether richer models add anything to a linear, exactly decomposable baseline when both receive the same information under a frozen chronological protocol.

The Temporal Fusion Transformer (TFT) [25] and N-HiTS [26] are the reference neural forecasters with covariates, and foundation models such as TimesFM and Chronos transfer representations learned from large time-series collections to new

tasks [27, 28]. Recent benchmarks have begun to map where these models help. The FETS benchmark evaluates foundation models against task-specific learners across 54 energy datasets [29]; zero-shot Chronos and TimesFM variants have been compared with tuned LSTMs and conventional models for electricity prices [30, 31]; and foundation models have been compared for monthly precipitation forecasting [32]. Comparative frameworks also stress the evaluation design itself, whether fixed-origin, expanding or rolling windows against linear and naive baselines [33], the effects of context length and horizon [34], or transfer and zero-shot strategies [35]. All of these operate on regularly sampled series. Industrial pelleting records are different. Consecutive completed operations have different masses, products, lines, start times and durations, so the prediction origin and the information available at that moment have to be fixed before any model can be compared [36], and random or shuffled partitions inflate apparent performance [37, 38, 39].

Explainable AI for energy systems has a growing literature [40, 41, 42], from frameworks that validate what data-driven power-curve models have learned [43] to permutation feature importance combined with conformal prediction to make explanations robust in predictive process monitoring [44]. Prediction intervals complete the picture. Conformal methods have been compared for electricity prices and solar power [45, 46] and used to calibrate multi-energy load forecasts [47], but their guarantees weaken under temporal dependence and drift [48, 49, 50], so coverage has to be read together with width and checked by subgroup [51, 52]. Digital twins are the natural destination of a validated baseline [53], provided its statistical component can be trusted.

Table 1 places the closest studies next to the present one. As far as we know, no previous work combines real operation-level records, chronological validation with a frozen test, an exactly decomposable baseline, residual-calibrated intervals checked by line, and a matched comparison with temporal and foundation models.

1.2. Objective and contributions

Our objective is to establish, on real operational records of three industrial feed-pelleting lines, whether an explainable, production-adjusted electricity baseline predicts the energy of a completed operation better than the line-specific SEC reference used in practice, and whether histories of completed operations, modern temporal models or pretrained foundation models add anything to it. The paper makes three contributions. The first is a production-adjusted baseline for heterogeneous operations validated with a frozen chronological test: seven estimators are compared on 800 chronological development targets, the selected Ridge regression is frozen together with its interval calibration, and it is then evaluated once on

Table 1: The closest published studies and what this one adds. "Operation-level" means one record per completed industrial operation with its production descriptors; "frozen test" means a model fixed before a later period is evaluated.

Study	Data and target	Production adjusted	Frozen chrono-logical test	Explainability / uncertainty	Temporal or foundation models
Kissock and Eger [12]	monthly plant energy; weather, production	yes	partly	linear terms / no	no
Granderson et al. [13]; Gallagher et al. [14]	interval meter data; buildings, industrial site	yes	yes	no / savings uncertainty	no
Sánchez et al. [16]	feed plant; motor efficiency, EnPIs	aggregated kWh/t	no	no / no	no
Ekwaro-Osire et al. [20]	≈1,100 pelleting batches, 67 inputs	yes	no (shuffled CV)	no / no	no
Wigger et al. [17]	company consumption series	no	forecast split	no / no	conventional forecasters
Panjapornpon et al. [21]	petrochemical signals; energy efficiency	process variables	no	post-hoc XAI / fault uncertainty	LSTM
Letzgus and Müller [43]	wind turbine SCADA; power curve	ambient conditions	yes	post-hoc XAI / no	no
FETS [29]; Miseta et al. [30]; Lin and Ti [31]	regular energy series; prices	covariates where available	yes	no / partly	foundation vs. task-specific
O'Connor et al. [45]; Nam et al. [46]	electricity prices; solar power	no	yes	no / conformal, global	conventional, Transformer
This study	**2,745 real pelleting operations; kWh per operation**	**yes**	**yes**	**exact additive / by line and week**	**TFT, N-HiTS, TimesFM 2.5, Chronos-2 on identical records**

798 later records against line-specific SEC. Previous feed-industry baselines are aggregated [16] or validated by shuffling [20]. The second is a matched test of operating histories, temporal models and foundation models on irregular industrial records: TFT, N-HiTS, TimesFM 2.5 and Chronos-2 receive exactly the same 16 completed same-line operations and the same current descriptors as the linear control, on 584 identical targets, with the information available at the origin fixed in advance. Existing foundation-model benchmarks concern regularly sampled series [29, 30, 31]. The third is record-level explainability and calibrated uncertainty that survive the chronological test: every estimate decomposes exactly into production, product, line and calendar terms, and residual-calibrated 90% and 95% intervals are assessed by line and by week, which is what allows an unusual operation to be told apart from an unusual product mix.

The validated scope is retrospective assessment at the closure of recorded operations. Prospective monitoring is a later application and would first require confirming which predictors are available at the intended origin. For a plant, the

result is a baseline that separates the ordinary variability of electricity use from deviations worth a closer look, and that can serve as the statistical component of an energy-management platform or digital twin. Section 2 describes the records, models and protocol, Section 3 the results, Section 4 their implications and limits, and Section 5 concludes.

2. Materials and methods

2.1. Study design

This is a retrospective, observational model-development study on the recorded operations of three industrial feed-pelleting lines. The quantity estimated is the expected electricity of a completed operation, not a continuously sampled power signal, and the observational unit is one row of the plant database. In the primary evaluation, models are developed and selected on temporally ordered validation periods; the selected model is then frozen, evaluated against operational reference baselines in a later test period, and assessed for predictive performance and interval coverage overall and by line. A set of complementary analyses looks at short histories of completed same-line operations with locally trained temporal architectures and pretrained time-series models, at line- and product-specific mass interactions, at a recent-residual correction, at rolling updates of the interval widths and at the numerical agreement between computational implementations. None of them replaces the frozen model. Figure 1 summarizes the workflow.

2.2. Industrial records and observational unit

The plant workbook holds 2,745 records and 32 named variables from three lines: 730 from Pellet 1, 1,148 from Pellet 2 and 867 from Pellet 3, with recorded stage starts from 2 January to 1 April 2025 and the last pelleting completion on 2 April. The plant information system generated each row as a consolidated record of one production operation: dosed mass, dosing-batch count, product classifications, stage timestamps, pelleting electrical energy (`KWH PELET`), steam-related variables and operational summaries. There is no verified production-order identifier linking every stage to a single physical batch and no separate meter history from which the energy value could be reconstructed, so a "recorded production operation" is one row of the workbook, and source-row identifiers are preserved throughout. Records with overlapping timestamps were kept when their identifiers differed, because overlap alone is no evidence of duplication; overlaps are characterized by line and treated as a data-quality descriptor.

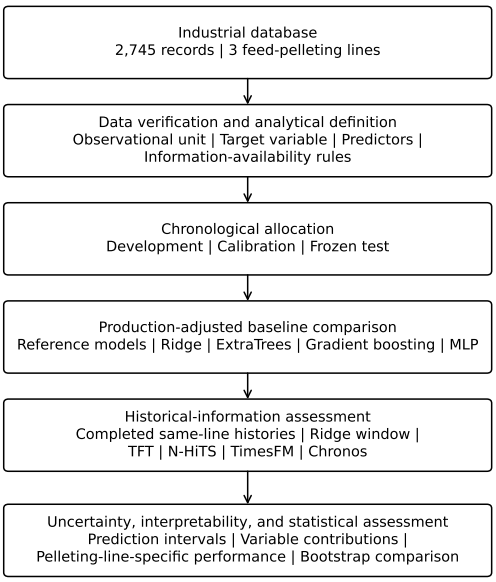


Figure 1: Methodological workflow. The primary evaluation comprises chronological model development, frozen test evaluation and line-specific assessment; the complementary analyses examine short-context temporal models, line- and product-specific interactions, recent-residual correction, rolling interval updates and computational diagnostics.

2.3. Target, predictors and preprocessing

The response y_i is `KWH PELET`, the electrical energy attributed to pelleting record i in kWh; it excludes the separately recorded electricity of milling and mixing and the thermal energy supplied as steam. Dosed mass m_i comes from `PESO DOS (kg)` converted to tonnes. Energy, a duration-based average-power proxy and specific energy consumption are defined as

$$E_i = y_i \ [\text{kWh}], \qquad P_i^{\text{avg}} = \frac{y_i}{\Delta_i} \ [\text{kW}], \qquad \text{SEC}_i = \frac{y_i}{m_i} \ [\text{kWh/t}], \tag{1}$$

where Δ_i is the recorded pelleting duration in hours. P_i^{avg} is derived, not measured; E_i is the modelling target and SEC is used only to build the reference baselines.

The analytical origin t_i is the earliest recorded start among milling, mixing and pelleting; it assigns records to partitions and decides whether a historical outcome was completed before a target. The predictors are dosed mass, dosing-batch count, pelleting line, product subfamily, presentation, bag/bulk category, origin hour and origin weekday (Table 2). With b_i the number of dosing batches, h_i the origin hour

and $d_i \in \{0, \ldots, 6\}$ the weekday from Monday, the numerical input vector is

$$x_i^{\text{num}} = \left(m_i, b_i, \sin \frac{2\pi h_i}{24}, \cos \frac{2\pi h_i}{24}, \sin \frac{2\pi d_i}{7}, \cos \frac{2\pi d_i}{7} \right). \tag{2}$$

The six numerical coordinates are standardized with training-data parameters only and the four categorical variables are one-hot encoded with training vocabularies, giving 22 columns in the Ridge design matrix; a level absent from a training period produces no active indicator. Grouped Ridge contributions therefore depend partly on this coding and are read as components of the fitted representation, not as invariant physical effects. The analysis is retrospective and conditional on the consolidated production characteristics of each operation: current-record durations, stage energies and the steam, temperature and current summaries are excluded as inputs because their availability at the origin cannot be established, and recorded pelleting completion is taken as the availability time of historical outcomes. A prospective implementation would have to restrict the model to descriptors verified as available at the intended origin.

2.4. Data integrity and temporal consistency

All 32 source fields were reconciled with the analytical table across the 2,745 records: the 87,840 cell-level comparisons satisfied the predefined numerical and whitespace tolerances. Three records (0.11%) with `KWH PELET=0` were excluded from supervised modelling, without imputation. Table 3 lists the main checks; the counts describe different characteristics of possibly the same records. Median pelleting durations are about 145, 96 and 123 minutes on Pellet 1, 2 and 3; milling completion falls after pelleting start in 528 records and mixing completion in 376. These patterns are characteristics of consolidated plant records rather than exclusion criteria. The workbook contains event-level summaries, not dense power trajectories or verified active/inactive labels; a separately supplied set of synthetic ten-second series was kept apart from the industrial evidence.

2.5. Chronological evaluation design

Records are assigned by origin to development (before 22 February 2025), calibration (22 February to 5 March) and testing (from 6 March); the last operation completes on 2 April (Figure 2). Development and calibration eligibility require a positive outcome, a valid duration, complete inputs and completion before the next boundary, which retains 1,542 of 1,556 development candidates and 380 of 391 calibration candidates (the 25 exclusions of Table 3); all 798 test candidates are retained, including the five operations longer than 24 hours.

Table 2: Model inputs: source fields, preprocessing, recorded support and conditions for prospective use. Original Spanish field names are retained for traceability; counts and ranges refer to all 2,745 source records.

Input / units	Source field and preprocessing	Interpretation and recorded support	Prospective use
Dosed mass m_i / t	`PESO DOS (kg)`; divided by 1,000	Material entering the operation; 2.69–36.78 t, median 14.98 t; the mass denominator used throughout	Possible if the mass is available at the intended origin
Dosing batches b_i / count	`N_BACHES_DOS`; numerical	One to ten, median five; describes how the mass is aggregated and is correlated with it; not a physical-cycle identifier	Possible when the batch plan or completed count is available
Pelleting line ℓ_i / category	`PELLT`; one-hot	Pellet 1, 2 and 3: 730, 1,148 and 867 records; the separate field `LINEA` holds product families	Possible when the line assignment is confirmed
Product subfamily c_i / category	`SUBLINEA`; one-hot	Nine labels spanning poultry, swine, cattle, equine and rabbit products; a classification, not a measured formulation	Possible when the product classification is known
Presentation p_i / category	`PRESENTACION`; one-hot	`PELETIZADO` (pelleted): 2,065; `CROMBELIZADO` (crumbled): 680; may also reflect product allocation	Possible when the presentation is known
Bag/bulk s_i / category	`DES_PRODUCTO_TIPO`; one-hot	`ENSAQUE` (bagged): 2,420; `GRANEL` (bulk): 325; the analytical alias `salida_programada` identifies output form, not a departure time	Possible when the output form is known
Origin hour / dimensionless pair	Earliest recorded stage start; sine and cosine	Two coordinates avoid a discontinuity between 23:59 and 00:00; seconds are not used	Available at or immediately after the origin
Origin weekday / dimensionless pair	Same origin; sine and cosine	Two coordinates represent recurring weekly associations, not operational state	Available at the origin

Table 3: Data-integrity and temporal-consistency checks.

Check	Count	Analytical treatment
Source records	2,745	Reconciled with the analytical table.
Records with zero pelleting energy	3	Excluded from supervised targets; not imputed.
Imputed auxiliary values	15	Retained in the source copy; affected variables were not model inputs.
Within-line interval overlaps	351	Retained as distinct source records and characterized by pelleting line.
Pellet 1 overlaps	75	Included in the line-level overlap count.
Pellet 2 overlaps	131	Included in the line-level overlap count.
Pellet 3 overlaps	145	Included in the line-level overlap count.
Test records exceeding 24 hours	5	Retained in the primary test population.
Pre-test eligibility exclusions	25	Fourteen development and eleven calibration candidates.

Model development uses four expanding training blocks with validation weeks beginning on 25 January, 1 February, 8 February and 15 February 2025: training sizes 705, 924, 1,121 and 1,334, validation sizes 208, 187, 205 and 200, i.e. 800 common targets over 28 origin dates. Training outcomes must be completed before each validation period starts, and validation outcomes before the week ends, so that the expanding blocks remain separated. Assessment thus follows the direction in which the model would be used and keeps future outcomes out of both model fitting and historical contexts [37, 38, 39, 54].

2.6. Reference baselines and current-input models

With ℓ_i the line of operation i, the line-specific SEC reference estimates one energy intensity per line from training totals:

$$\widehat{s}_\ell = \frac{\sum_{i\in\mathcal{T}:\ell_i=\ell} y_i}{\sum_{i\in\mathcal{T}:\ell_i=\ell} m_i}, \qquad \widehat{y}_i^{\mathrm{SEC}} = m_i \widehat{s}_{\ell_i}. \tag{3}$$

A line-specific median-energy reference and a recent-SEC reference (median of the last five completed same-line energy-to-mass ratios, updated as outcomes arrive during validation) complete the set of operational references. With z_i the transformed current-record inputs, Ridge regression estimates an unpenalized

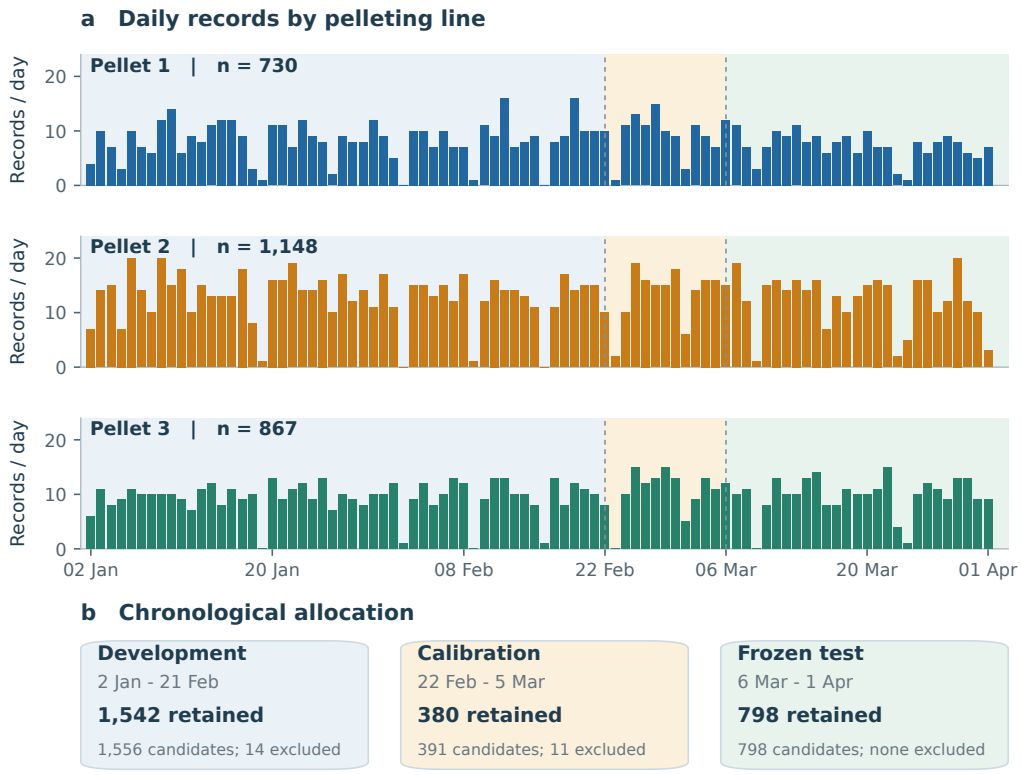


Figure 2: Chronological allocation by pelleting line and partition. (a) Daily record counts for the three lines; background shading marks the development, calibration and frozen-test partitions, and dashed lines mark the development–calibration boundary on 22 February 2025 and the start of testing on 6 March 2025. Each record is counted by its earliest recorded stage start. (b) Dates, candidate records, exclusions and retained sample sizes of each partition.

intercept and regularized coefficients,

$$(\widehat{b}, \widehat{\beta}) = \arg\min_{b,\beta} \left\{ \sum_{i \in \mathcal{T}} (y_i - b - z_i^\top \beta)^2 + 10 \|\beta\|_2^2 \right\}, \qquad \widehat{y}_i = \max(0, \widehat{b} + z_i^\top \widehat{\beta}), \tag{4}$$

where the non-negative truncation prevents negative energy estimates without imposing a process-energy balance. ExtraTrees (250 trees, minimum leaf size five), histogram gradient boosting (200 iterations, seven leaves, learning rate 0.05, minimum leaf size 20, regularization 10) and a multilayer perceptron (hidden layers of 32 and 16 units, Adam, regularization 0.1, at most 800 iterations, seeds 11, 29 and 47 averaged) use the same inputs with configurations held constant across

validation blocks; the tree methods use seed 2026. We evaluated these particular implementations; no attempt was made to optimize each family exhaustively. When Ridge and gradient boosting gave practically the same development MAE, Ridge was chosen for parsimony, stability and record-level traceability, and the selected model and its interval calibration were frozen before any test record was evaluated.

For interpretation, the 22 design columns are assigned to eight groups (dosed mass, batch count, hour, weekday, line, subfamily, presentation, bag/bulk), so that the untruncated estimate and the contribution of group g are

$$f_i^{\mathrm{Ridge}} = \widehat{b} + \sum_{g \in \mathcal{G}} C_{ig}, \qquad C_{ig} = \sum_{k \in g} z_{ik} \widehat{\beta}_k, \qquad \widehat{y}_i = \max(0, f_i^{\mathrm{Ridge}}). \tag{5}$$

The grouped contributions preserve the arithmetic of the fitted model; they describe how the encoded variables build an estimate, not separate physical energy uses or causal effects [55, 56, 57].

2.7. Short-context temporal models on irregular operational events

The temporal comparison asks a simple question: do histories of completed same-line operations add information beyond the current-record covariates? Each eligible target receives the 16 most recent completed records from the same line and validation interval,

$$\mathcal{H}_i^{(16)} = \mathrm{last}_{16,(a_j,j)}\{j : \ell_j = \ell_i,\ t_j \in \mathcal{P},\ a_j < t_i,\ a_j < \sup \mathcal{P}\}, \tag{6}$$

where a_j is the recorded completion time and ties are broken by source row; an operation that started earlier but was not complete at the target origin cannot enter the context. The window retains 584 of the 800 development targets (149, 257 and 178 from Pellet 1, 2 and 3 over 21 origin dates). Contexts restart at the weekly validation boundaries; completed outcomes may inform later targets within the week but never update fitted parameters. Figure 3 shows the design and one observed context.

Two Ridge controls separate historical information from model capacity: one uses the current covariates only, the other appends the 16 historical covariate vectors and their energies, keeping the penalty of 10. TFT and N-HiTS receive the same information through their NeuralForecast implementations [58, 59]; TFT uses hidden size 32, four heads and dropout 0.1, N-HiTS three blocks of two 64-unit layers with pooling 2, 2, 1. Each uses seeds 11, 29 and 47 with an inner validation week before each outer week for epoch selection (Adam at 10^{-3}, weight decay 10^{-4}, batch 64, gradient clipping 1, at most 40 epochs, patience 8); each seed is refitted on

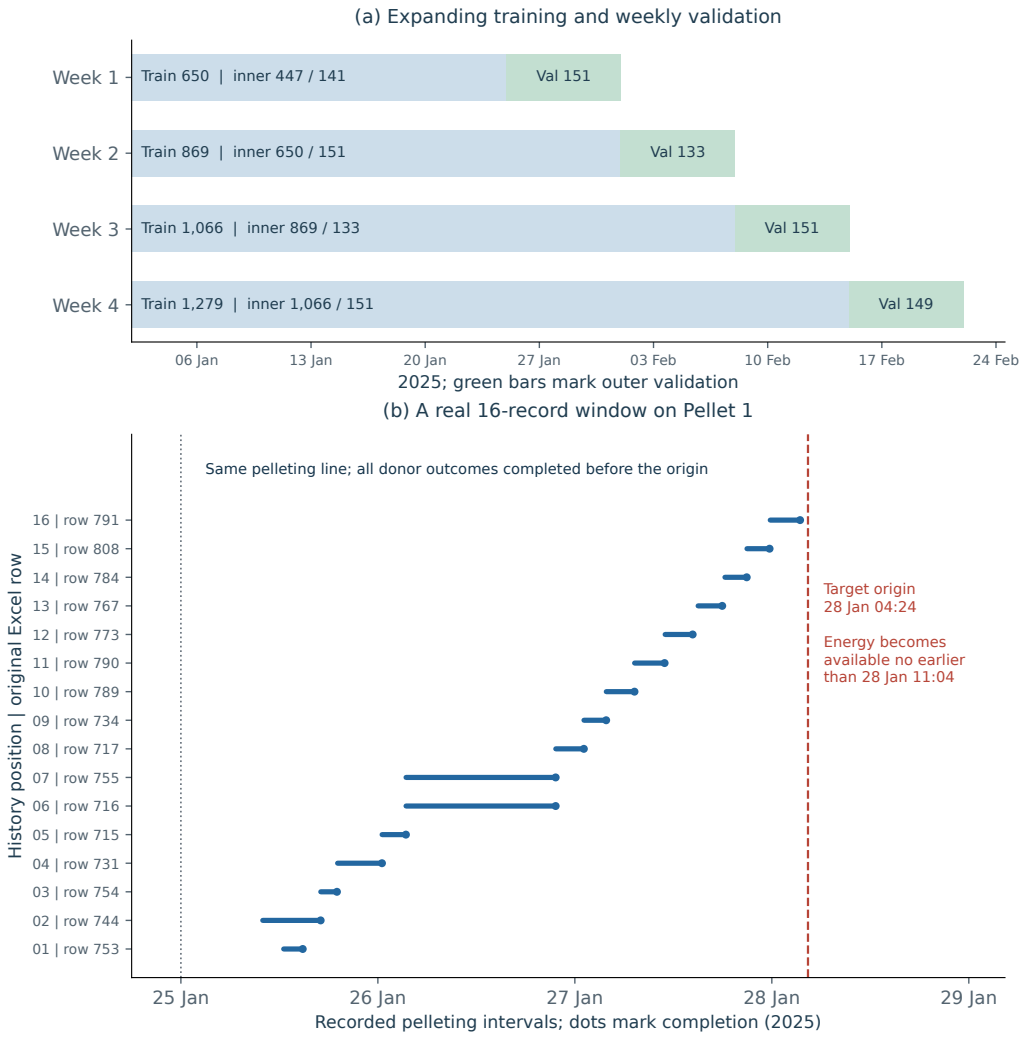


Figure 3: Short-context temporal evaluation. (a) Expanding training blocks and weekly validation blocks; green bars mark the outer validation weeks, and each training bar reports the earlier training / preceding-week validation counts used for epoch selection. (b) An observed context of 16 completed same-line operations for target row 823 on Pellet 1, ordered by completion; segments span recorded pelleting start and finish, dots mark completion, and every donor outcome precedes the target origin on 28 January at 04:24.

the full outer-training block for its selected epochs and the three clipped predictions are averaged. TimesFM 2.5 runs zero-shot with its frozen 200M checkpoint, alone and with its XReg adapter (ridge 10, one call per origin so that no query pools labels from another query's future) [27, 60, 61]; Chronos-2 runs with energy-only and native covariate inputs, median forecast, cross-query learning disabled [62, 63]. All variants are evaluated on the same 584 targets, so the comparison concerns these short-context implementations on irregular pelleting events, not the model families in general [64, 65].

2.8. *Post-hoc extensions*

Three extensions turn patterns noticed during development into hypotheses for later periods. H1 lets the mass association vary by line, and H2 by product subfamily, through the untruncated predictor

$$f_i = b + z_i^\top \beta + z_{m,i} \sum_{c \in C} \gamma_c \mathbf{1}(g_i = c), \tag{7}$$

with grouping variable g_i, the same penalty and training-only transformations. H3 corrects the frozen estimate with up to 20 recent same-line residuals $e_j = y_j - \widehat{y}_j^{\text{Ridge}}$ of completed operations, ordered by completion and source row; with $n_i = |\mathcal{H}_i|$,

$$\delta_i = \begin{cases} \dfrac{n_i}{n_i + 10} \dfrac{1}{n_i} \displaystyle\sum_{j \in \mathcal{H}_i} e_j, & n_i \geq 5, \\ 0, & n_i < 5, \end{cases} \qquad \widehat{y}_i^{\text{H3}} = \max(0, \widehat{y}_i^{\text{Ridge}} + \delta_i). \tag{8}$$

H1–H3 are exploratory development analyses; they do not replace the frozen model.

2.9. *Residual intervals and performance metrics*

Absolute errors on the 380 calibration records calibrate the intervals of the frozen model. For miscoverage level α,

$$q_\alpha = r_{(\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil)}, \qquad r_j = |y_j - \widehat{y}_j|, \qquad I_i^{1-\alpha} = [\max(0, \widehat{y}_i - q_\alpha), \widehat{y}_i + q_\alpha], \tag{9}$$

giving half-widths of 93.01 and 132.96 kWh at the 90% and 95% levels. They are called residual intervals because the industrial sequence is not assumed exchangeable; they are judged empirically by coverage, mean width and interval score, globally and by line. A post-hoc rolling policy recomputes the quantile from the latest 200 available absolute errors, starting from the calibration residuals and admitting a test residual only once the completion of its operation precedes the

target origin ($a_j < t_i$); it is an exploratory comparator, not a validated improvement [66, 67, 68].

Point performance is measured by

$$\mathrm{MAE} = N^{-1} \sum_i |\widehat{y}_i - y_i|, \qquad \mathrm{RMSE} = \sqrt{N^{-1} \sum_i (\widehat{y}_i - y_i)^2}, \tag{10}$$

$$\mathrm{Bias} = N^{-1} \sum_i (\widehat{y}_i - y_i), \qquad \mathrm{WAPE} = \frac{\sum_i |\widehat{y}_i - y_i|}{\sum_i y_i}, \tag{11}$$

with MAE as the selection criterion [69]. Intervals are assessed by coverage, mean width and the interval score [52],

$$\mathrm{IS}_\alpha(L, U; y) = (U - L) + \frac{2}{\alpha}(L - y)\mathbf{1}(y < L) + \frac{2}{\alpha}(y - U)\mathbf{1}(y > U), \tag{12}$$

and each completed operation yields a signed excess $r_i^{\mathrm{excess}} = y_i - \widehat{y}_i$ and an out-of-interval flag $B_i = \mathbf{1}\{y_i < L_i \text{ or } y_i > U_i\}$ that mark candidates for review without identifying the physical cause. Paired differences use a circular block bootstrap over observed origin dates (blocks of three dates, all lines of a date together; 3,000 replicates for the test comparison, 5,000 for the exploratory ones), with Bonferroni-adjusted 98.33% intervals for H1–H3 and 99.29% intervals for the seven temporal contrasts alongside the marginal 95% intervals.

2.10. Computational implementation

Ridge and the native JAX XReg solver ran on CPU; neural computations used float32 on an NVIDIA RTX PRO 5000 Blackwell GPU (48 GB) with TF32 disabled. TFT and N-HiTS were trained locally, whereas TimesFM and Chronos weights remained frozen. CPU–GPU numerical comparisons are reported in Appendix Appendix A.

3. Results

3.1. Production variability across lines and periods

Figure 4 summarizes production and electricity on the three lines: panels (a)–(c) contain the 2,742 records with positive pelleting energy, and panels (d)–(f) the mass, dosing-batch and product-family distributions of all 2,745 records. Electricity rises with dosed mass, yet records of similar mass remain widely dispersed, and the distributions of mass, batch count and product family differ among lines and

partitions. Both observations argue for a production-adjusted baseline rather than a single plant-wide intensity. Line-level differences in consumption have to be read together with the corresponding production allocation: they are not controlled comparisons and say nothing about intrinsic mechanical efficiency.

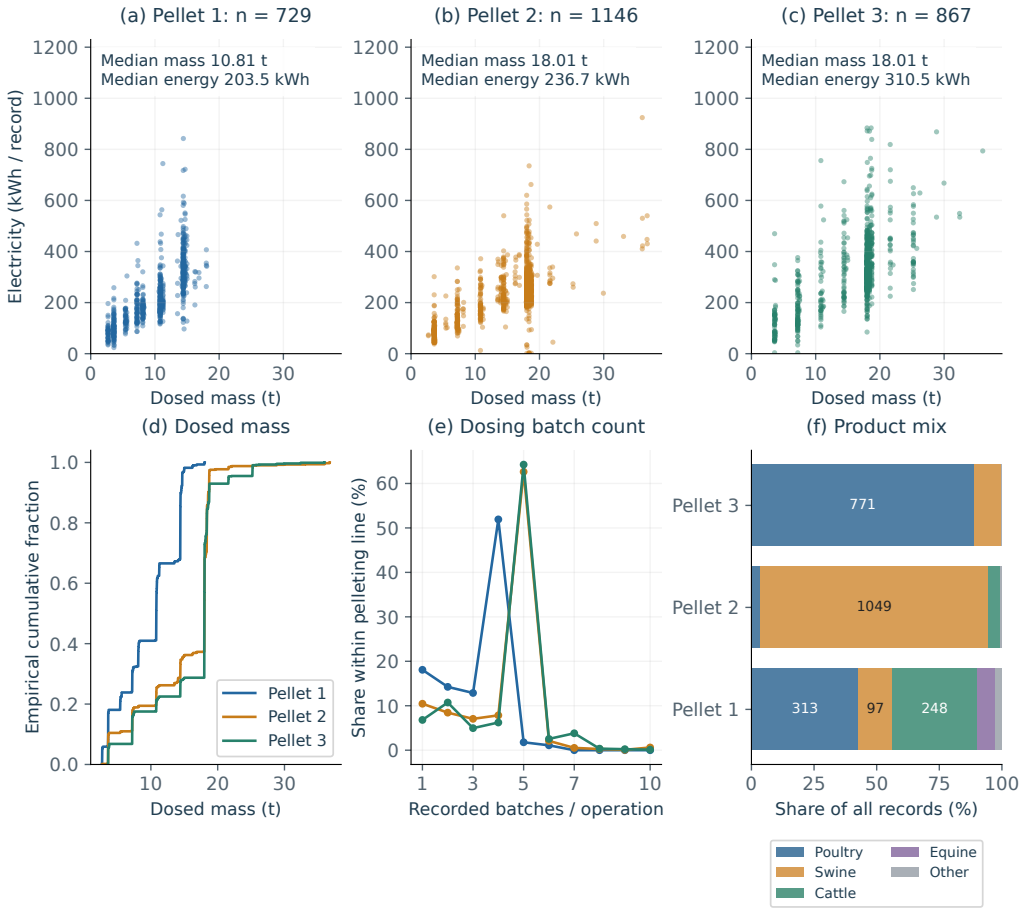


Figure 4: Production and electricity across pelleting lines. (a–c) Dosed mass and pelleting electricity for the 2,742 positive-energy records, with line-specific medians annotated. (d–f) Mass distributions, dosing-batch frequencies and product-family mix of all 2,745 source records; values inside the product bars are record counts. Points are individual operations and are not uniformly spaced in time.

3.2. Baseline selection and frozen chronological test

On the 800 development targets (Table 4), Ridge and histogram gradient boosting were practically tied at 51.48 and 51.50 kWh MAE, with ExtraTrees close

behind at 52.31 kWh. The MLP and the three references had larger errors, and the recent-SEC estimator did worse than the static line-specific SEC even though it used completed validation outcomes. Ridge was selected because it matched the best accuracy with far less structural complexity and with an exact additive decomposition, not because it was shown to beat gradient boosting.

Table 4: Chronological development performance on the same 800 records. The recent-SEC estimator was updated as completed outcomes became available during validation; all other models kept fixed parameters within each validation block.

Method	MAE (kWh)	RMSE (kWh)	Bias (kWh)	WAPE (%)
Ridge	51.48	72.31	−3.14	19.88
Gradient boosting	51.50	73.46	−5.72	19.88
ExtraTrees	52.31	75.37	−4.89	20.20
MLP, three-seed mean	56.64	81.36	−1.47	21.87
Line-specific SEC	59.69	83.92	−6.59	23.05
Recent SEC, last five records	73.51	109.55	4.29	28.38
Line-specific median	82.61	112.78	−14.09	31.89

Fitted on the 1,542 eligible development records and frozen, Ridge reduced test MAE from 61.29 to 52.45 kWh (Table 5): a paired reduction of 8.84 kWh per record, or 14.42%, with a 95% block-bootstrap interval of [6.08, 11.50] kWh. RMSE fell from 92.05 to 80.42 kWh and global WAPE from 22.34% to 19.11%. These are reductions in prediction error, not in plant electricity use. Removing the five test records longer than 24 hours gave 52.32 kWh for Ridge and 61.15 kWh for SEC, so those records do not explain the advantage.

The improvement appeared on every line (Figure 5), but the errors that remained were uneven. Pellet 2 and Pellet 3 were the most consistent, with Ridge MAE of 43.36 and 53.36 kWh. Pellet 1 also benefited, its MAE falling from 72.31 to 66.37 kWh and its bias from −64.03 to −53.85 kWh, but the underestimation that remains means the frozen model was not operationally calibrated for that line during the test period. The shortfall is local and does not invalidate the baseline; what it does mean is that one alerting criterion should not be applied across lines without line-specific calibration.

Table 5: Frozen chronological test performance overall and by pelleting line. Global WAPE is 19.11% for Ridge and 22.34% for line-specific SEC.

Population	Method	N	MAE (kWh)	RMSE (kWh)	Bias (kWh)
All lines	Ridge	798	52.45	80.42	−21.89
All lines	Line-specific SEC	798	61.29	92.05	−24.57
Pellet 1	Ridge	200	66.37	105.14	−53.85
Pellet 1	Line-specific SEC	200	72.31	113.90	−64.03
Pellet 2	Ridge	333	43.36	61.26	−10.85
Pellet 2	Line-specific SEC	333	49.29	73.24	−8.66
Pellet 3	Ridge	265	53.36	80.10	−11.63
Pellet 3	Line-specific SEC	265	68.05	94.79	−14.79

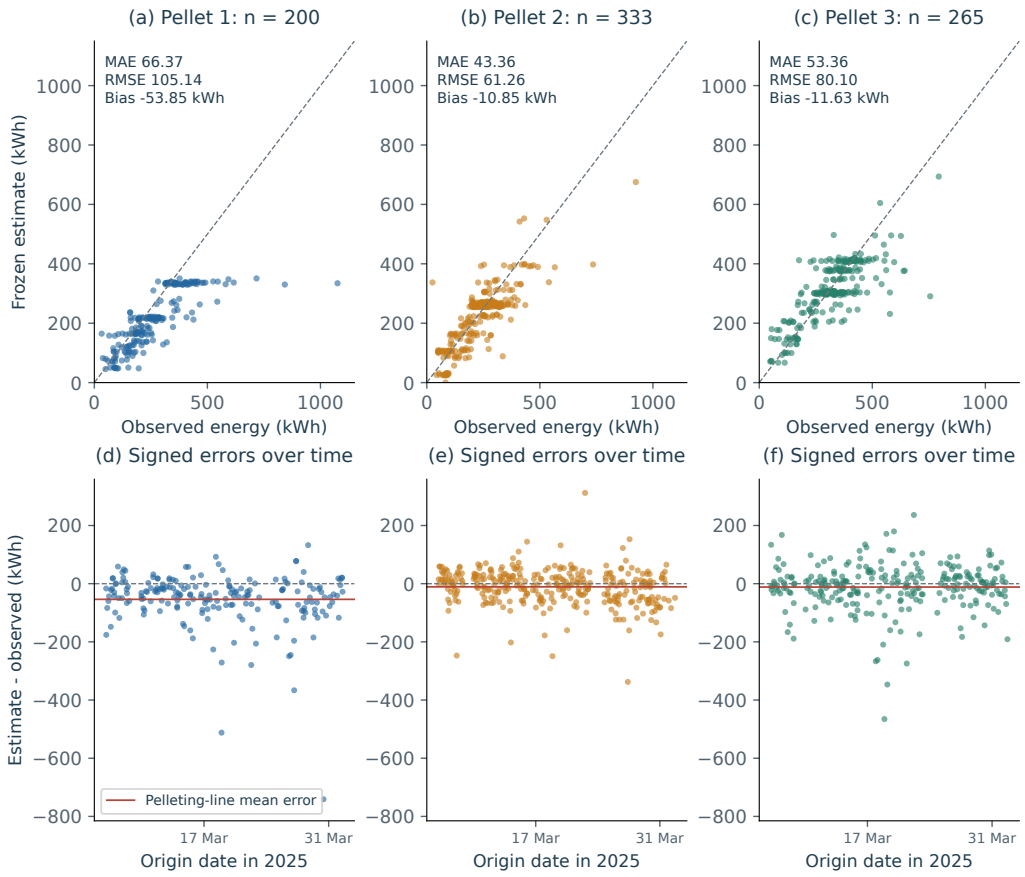


Figure 5: Frozen Ridge performance in the chronological test. (a–c) Observed versus predicted electricity on identical axes, with the equality line dashed and MAE, RMSE and bias in kWh. (d–f) Signed errors in chronological order within each line, with zero dashed and the mean signed error of each line in red. The 200, 333 and 265 records of Pellet 1, 2 and 3 make up all 798 test observations.

3.3. Predictive value of completed operating histories

The short-context comparison uses the 584 development records for which 16 completed same-line operations were available within the weekly interval; the other 216 did not meet the requirement, so these results are not comparable with the 800-record ranking. Current-input Ridge reached 53.55 kWh MAE and TFT 53.65 kWh (Table 6). The paired difference (Ridge minus TFT) was −0.11 kWh with a 95% interval of [−2.20, 2.32] kWh, which includes effects in both directions and gives no support for an improvement from TFT. Giving Ridge the full 16-record history actually hurt, raising its MAE to 63.95 kWh, and N-HiTS reached 65.28 kWh. Chronos-2 did better with covariates than with energy alone, but both versions stayed well above the Ridge control, as did the energy-only TimesFM variant (Figure 6).

Table 6: Short-context model performance on the 584 matched development records. TFT and N-HiTS are three-seed prediction means; foundation-model weights remain frozen.

Model group	Method	MAE	RMSE	Bias	WAPE (%)
Ridge controls	Current inputs	53.55	77.01	−4.72	20.37
	Full 16-record window	63.95	85.95	−1.42	24.32
Local temporal	TFT	53.65	79.11	−11.23	20.41
	N-HiTS	65.28	92.56	−3.05	24.83
Frozen pretrained	Chronos-2 + covariates	85.53	120.82	−8.11	32.53
	TimesFM 2.5, energy only	96.40	130.72	−9.22	36.67
	Chronos-2, energy only	100.02	136.47	−11.60	38.04
	TimesFM 2.5 + XReg	112.12	644.61	28.03	42.65

MAE, RMSE and bias are in kWh per record.

The TimesFM XReg variant produced three extreme Pellet 3 predictions of about 2,753, 9,239 and 12,203 kWh for observed values of 185.4, 142.4 and 70.9 kWh: in those contexts the dosed mass of the 16 donors was almost constant, so the standardized current mass lay roughly 1,100–1,467 historical standard deviations from the local mean and the adapter extrapolated accordingly. An independent float64 reconstruction reproduced the behaviour, and the three predictions are retained in every metric. The diagnosis concerns this short-context XReg pipeline, not TimesFM with longer contexts or other scaling policies. Under the tested representation, historical inputs did not improve prediction over current-record covariates; results by week, line and seed and the paired intervals of every contrast are given in Supplementary Tables S12–S16 and Figs. S3–S4.

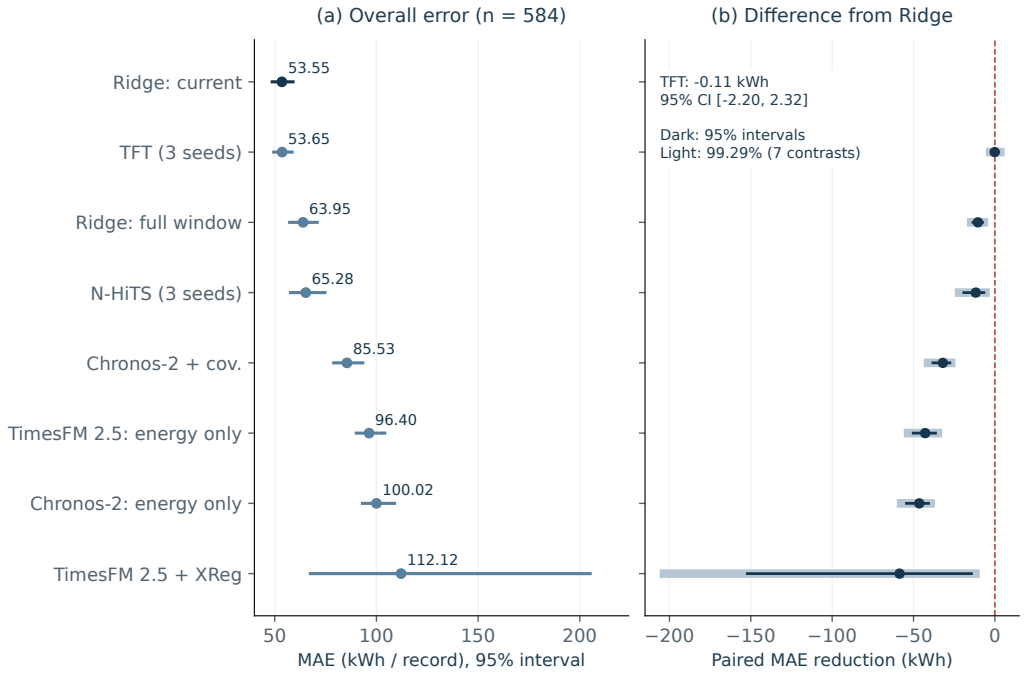


Figure 6: Short-context comparison on the 584 matched development records. (a) MAE with 95% block-bootstrap intervals. (b) Paired MAE reduction relative to current-input Ridge, so positive values favour the alternative; pale intervals are adjusted for seven comparisons and dark intervals are unadjusted 95% intervals. The wide TimesFM XReg interval is shown in full.

3.4. Additive decomposition of the fitted predictor

Figure 7 shows what the baseline returns for one operation: panel (a) the transformation of recorded production conditions into an expected electricity value, a residual interval and a signed deviation, and panel (b) the output for source row 2080, a Pellet 2 operation with 18.002 t in five dosing batches. The model estimated 255.90 kWh; the recorded value was 267.30 kWh, a residual of 11.40 kWh inside the fixed 95% interval of [122.94, 388.85] kWh. The record was chosen because its energy is close to the median test value, not because its error is typical.

Applying Equation 5 to every frozen estimate reproduced all 798 archived test values to within 1.2×10^{-13} kWh, and none required clipping. Figure 8 separates the fitted coefficients from record-specific contributions: panel (a) gives the numerical coefficients, (b) the categorical coefficients under the fitted coding, (c) the distributions of the grouped contributions and (d) the decomposition of source row 2080, whose intercept and eight grouped contributions sum exactly to

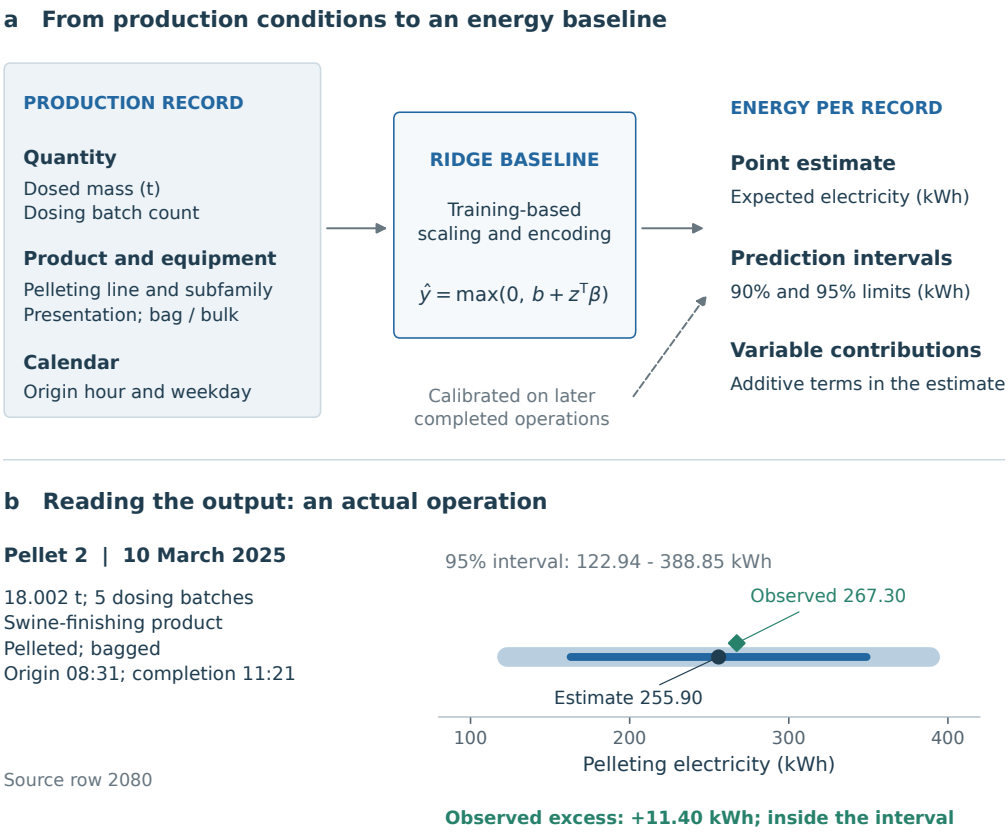


Figure 7: Production-adjusted baseline output. (a) Recorded production conditions are transformed into an expected electricity value, residual intervals and a signed deviation; residuals from the later calibration period set the interval limits. (b) A real Pellet 2 operation (source row 2080): the light segment is the fixed 95% interval, the dark segment the 90% interval, the circle the estimate and the diamond the observed energy. An excess above the estimate is a model residual, not a measured saving or a confirmed fault.

255.90 kWh. After undoing the standardization, the mass coefficient is 13.30 kWh per additional tonne and the batch coefficient 9.52 kWh per additional batch, conditional on the other encoded predictors (all 22 coefficients and conversion factors are in Supplementary Tables S6–S7).

These values are coefficients of a fitted linear predictor, not separately identified physical requirements: mass and batch count are correlated, and line and product allocation were not randomized. The intercept of 288.09 kWh is the algebraic reference of the coding rather than a standby consumption, the categorical coef-

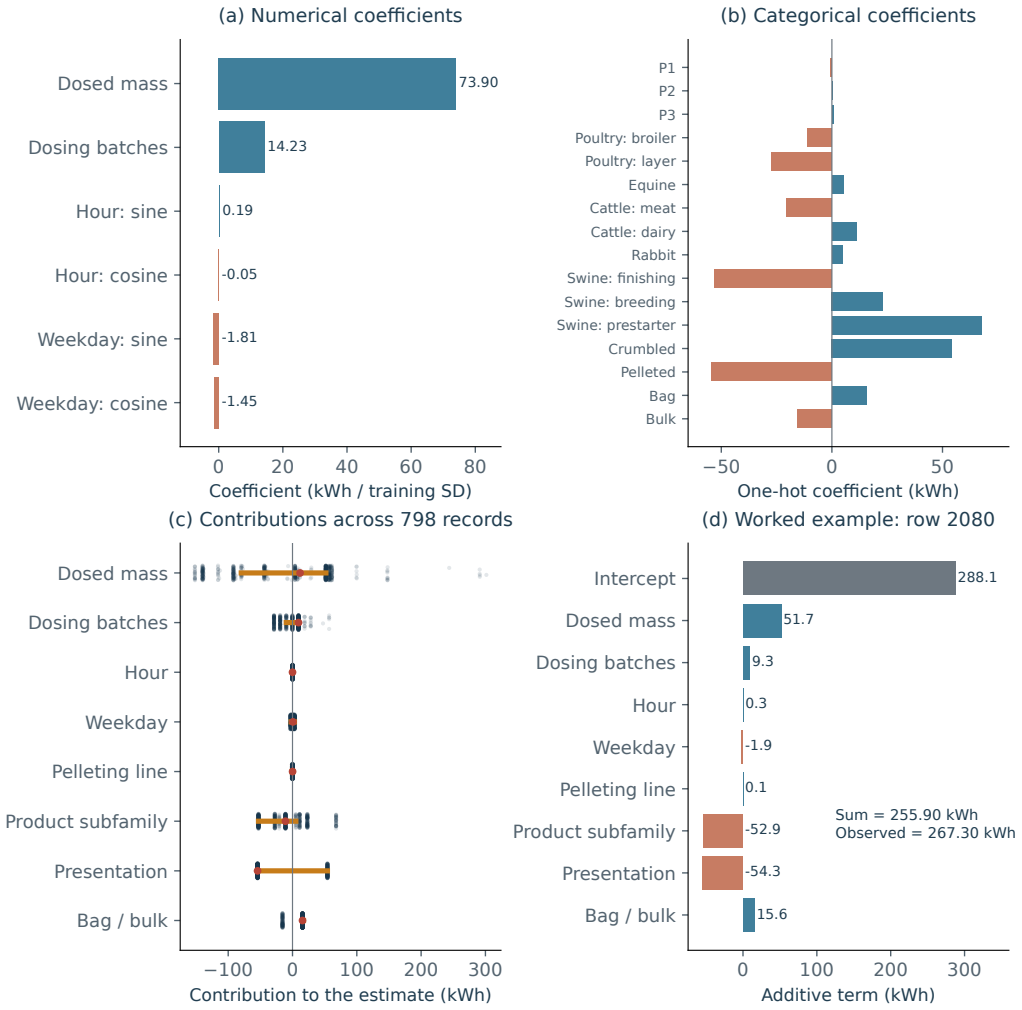


Figure 8: Additive decomposition of the frozen Ridge predictor. (a) Numerical coefficients in original input units. (b) All 16 categorical coefficients under the fitted coding. (c) Grouped contributions for every test record, with medians in red and interquartile ranges in orange. (d) Arithmetic decomposition of source row 2080. The reference combines standardized numerical means with all-zero categorical indicators and need not describe a feasible operating condition; contributions are conditional model terms, not causal effects.

ficients depend on the one-hot representation, and small line coefficients do not demonstrate equal intrinsic efficiency. What the exact equality does provide is an audit trail from every estimate back to the recorded inputs that produced it.

3.5. Post-hoc extensions

On the 800 development records (Table 7), letting the mass association vary by product subfamily reduced MAE by 1.44 kWh (2.80%) and the recent-residual correction by 1.71 kWh (3.33%), both with multiplicity-adjusted intervals above zero; the line-dependent mass interaction reduced MAE by only 0.22 kWh, with an interval crossing zero. The gains varied across lines and weeks (Supplementary Fig. S1 and Tables S8–S9). Because H1–H3 were motivated by development patterns, they are hypothesis-generating and need a new chronological period; they do not replace the frozen specification.

Table 7: Post-hoc exploratory point-model extensions on the 800 development records. Positive reductions favour the extension; the adjusted intervals cover these three comparisons only.

Variant	MAE	MAE reduction	Adjusted 98.33% interval
H1: mass by pelleting line	51.26	0.22	[−0.08, 0.58]
H2: mass by subfamily	50.04	1.44	[0.55, 2.48]
H3: recent residuals	49.77	1.71	[0.24, 3.16]

All quantities are in kWh per record.

3.6. Residual intervals and records flagged for review

The fixed 90% and 95% intervals covered 85.96% and 93.23% of the 798 test records (Figure 9) with mean widths of 183.63 and 258.85 kWh; their block-bootstrap 95% coverage intervals, [81.86, 90.24]% and [90.53, 95.87]%, make global coverage broadly compatible with the nominal levels. Line-specific coverage differed: at the 95% level it was 90.00% on Pellet 1 (20 records outside), 96.40% on Pellet 2 (12) and 91.70% on Pellet 3 (22). In total 54 records fell outside the 95% intervals, 45 above and nine below, which is about 6.77 flagged records per 100 operations. They are candidates for technical review because the calibrated baseline did not represent their energy well; without plant evidence they should not be called anomalies, inefficient operations or avoidable consumption.

The post-hoc rolling policy, which updates the quantiles from the most recent 200 completed residuals, raised coverage to 89.47% and 94.49% but widened the intervals by 19.66% and 18.85% while improving the interval score by only

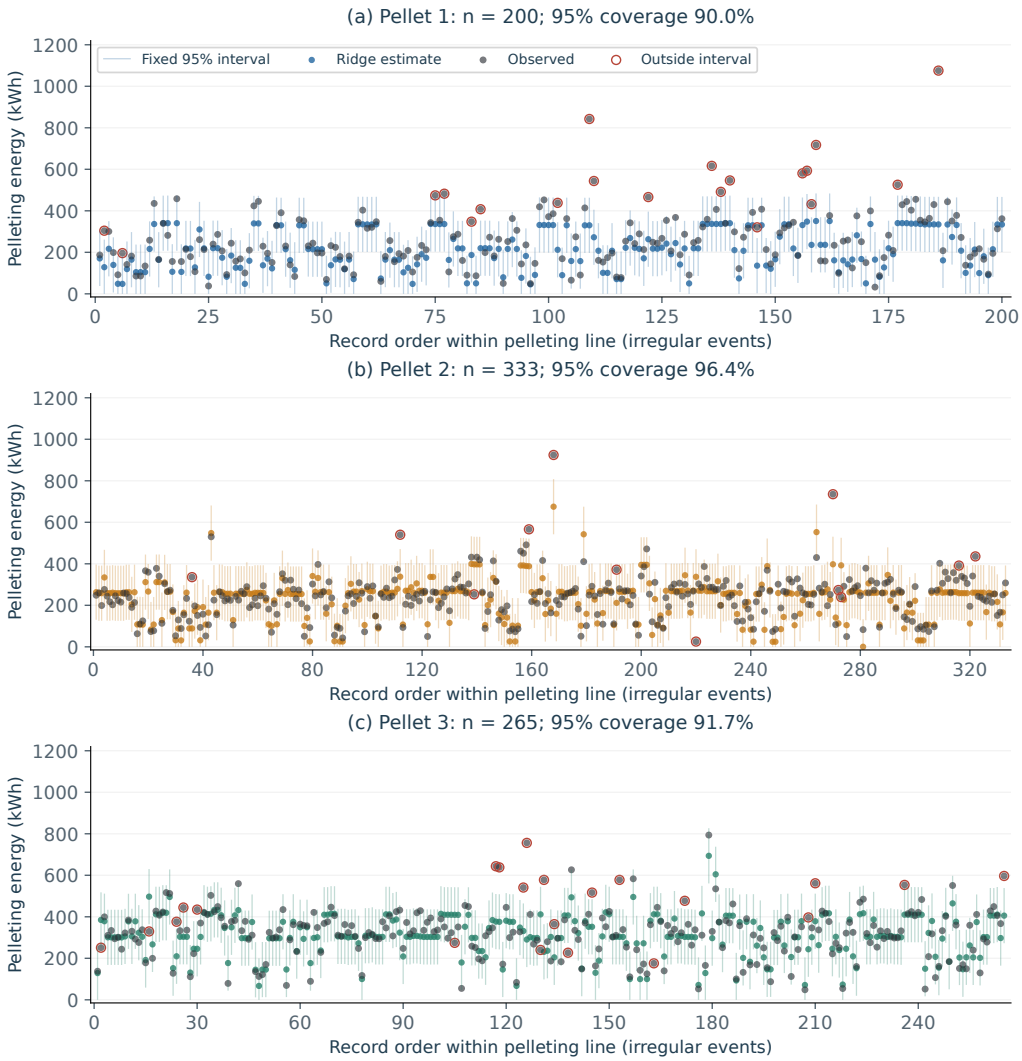


Figure 9: Frozen Ridge estimates and fixed 95% residual intervals for every test record, in chronological order within each pelleting line. Black points are observed electricity, coloured points the frozen estimate and vertical lines the intervals; red circles mark the 54 outcomes outside their intervals. Horizontal position is record order, so equal spacing does not imply equal elapsed time.

1.12% and 0.23% (Table 8). Most of the additional coverage therefore came from width; the number of records outside the 95% interval fell from 54 to 44. Because the policy was defined after the test results had been inspected, it is a calibration diagnostic rather than a validated replacement for the fixed intervals. Coverage, width and interval score by line and week are given in Supplementary Tables S10–S11 and Fig. S2.

Table 8: Fixed and post-hoc rolling residual intervals on the same 798 frozen point predictions. The rolling policy has not been validated on an independent period.

Policy	Nominal level	Coverage (%)	Mean width	Interval score
Fixed	90%	85.96	183.63	381.79
Rolling, 200 residuals	90%	89.47	219.74	377.51
Fixed	95%	93.23	258.85	497.64
Rolling, 200 residuals	95%	94.49	307.64	496.48

Width and interval score are in kWh per record.

4. Discussion

4.1. Production adjustment and what it changed on each line

During development, Ridge produced lower prediction errors than the line-specific median, the recent-SEC estimator, the MLP, ExtraTrees and line-specific SEC, and performed like histogram gradient boosting; in the frozen chronological test its MAE was 14.42% below line-specific SEC. This is in line with the energy-baseline literature, which has long recommended adjusting consumption for the relevant operating conditions instead of relying on aggregate or intensity-based indicators [12, 13, 14, 15], and it extends the pelleting studies that link energy demand to production volume, formulation, conditioning, equipment and operating conditions [10, 7, 8, 9, 11]. What is added is a chronologically evaluated baseline that combines production mass, product characteristics and line identity for heterogeneous completed operations, which is the normalization that ISO 50006 and the IPMVP expect of an energy baseline [2, 3]. Two qualifications should be kept in view. The improvement is in prediction accuracy, not in actual energy use, and it is not an estimate of achievable savings. And Ridge was chosen for its balance of accuracy, parsimony and traceability, not because it was shown to beat nonlinear models in general.

Production adjustment helped on all three lines, but not equally. Pellet 2 and Pellet 3 showed the most consistent agreement between predictions and observations and are the strongest basis for a prospective validation. Pellet 1 also improved on line-specific SEC, yet its remaining negative bias means the frozen model was not operationally calibrated for that line during the test period. That is a local requirement, not a failure of the common framework, and it illustrates how an aggregate score can hide a line-specific deficiency. Differences among lines may reflect equipment configuration, production allocation, maintenance, metering or unrecorded operating conditions, and should not be read as differences in intrinsic energy efficiency.

4.2. Operating histories, temporal models and foundation models

Under the tested representation (16 completed same-line operations within the validation week), histories did not improve on current-record covariates. One plausible reason is that dosed mass, product characteristics and line identity already capture most of the repeatable structure; another is that irregular intervals between operations weaken whatever meaning sequential proximity might have. The conclusion is confined to the evaluated short-context representation and does not contradict the value of temporal models where sequences carry informative dependence [25, 26]; likewise, the TimesFM and Chronos results characterize the tested interfaces, not the general capabilities of foundation models [27, 28, 62]. The XReg extrapolation and the device-dependent normalization sensitivity of Chronos-2 (Appendix Appendix A) do carry a more general lesson: covariate adaptation and normalization shape the final industrial prediction as much as the pretrained weights do, and the fact that a model accepts the available covariates says nothing about whether its pipeline is stable for this kind of record. A fairer assessment of temporal information in pelleting would need longer contexts, explicit elapsed time, operating-state information, maintenance events and regularly sampled measurements [64, 65].

4.3. Reading the decomposition and the intervals

Ridge gives an exact additive decomposition of its fitted linear predictor, so each estimate can be traced back to its inputs directly and reproducibly. Its coefficients and record-specific contributions nevertheless depend on scaling, categorical coding, reference categories, feature construction and correlations among predictors [55, 56, 57]: they describe the fitted prediction function, do not identify causal effects, and would change under alternative coding choices. Line and product terms may also absorb unrecorded differences in formulation, equipment

condition, production allocation or operating practice. The small exploratory gains from product-dependent mass effects and residual correction point to where the next refinement may come from, but they are hypotheses for a new period rather than reasons to replace the frozen model on the strength of a test that has already been examined.

The residual intervals made a related point. Satisfactory marginal coverage did not guarantee comparable calibration on each line [50, 48, 49]: Pellet 2 had the most stable coverage, whereas Pellet 1 and, to a lesser extent, Pellet 3 need further assessment. Rolling calibration raised coverage mainly by widening the intervals, which is a reminder that coverage has to be read together with interval width, bias and alert frequency [51, 52]. An interval exceedance is therefore a candidate for technical review. It is not, by itself, a confirmed anomaly, an inefficiency, an equipment failure or an energy-saving opportunity.

4.4. Scope of use, limitations and digital-twin integration

The evidence supports retrospective assessment after each recorded operation closes: an expected electricity value, an empirically calibrated residual interval and a traceable deviation that identifies candidates for review. Prospective supervisory alerts would require validation over a new consecutive production period with prespecified line-level criteria and, for Pellet 1, additional recalibration or predictors. Automatic control and modification of process set-points remain outside the evidence, because an observational model does not estimate causal responses. A digital-twin application should therefore begin with validated monitoring and decision support [53], storing the inputs and model version of each issued estimate and attaching the measured outcome when it becomes available, so that an earlier forecast and a later production-adjusted assessment remain distinguishable.

The study draws its strength from real industrial records, chronological separation, operational comparators, line-specific diagnostics, residual intervals and a reproducible decomposition. Its transferability is limited by the absence of a verified production-order identifier, by retained temporal overlaps, by unverified meter boundaries and by undocumented predictor-availability times. The data cover three lines of one plant over about three months, use product categories rather than complete formulations, and lack detailed maintenance and operating-state variables. The intervals mainly assess marginal coverage, several extensions were post hoc, and the operational records cannot be released publicly, although they are available from the corresponding author on reasonable request. The contribution therefore lies in the evaluation framework, with its production adjustment, chronological testing, line-specific assessment, uncertainty quantification and algebraic traceability,

rather than in a fitted model that could be deployed elsewhere as it stands.

5. Conclusions

On 798 real pelleting records evaluated after the model had been frozen, a production-adjusted Ridge baseline reduced MAE by 14.42% relative to the line-specific SEC reference that plants use in practice, on all three lines, while every estimate remained exactly decomposable into its inputs. The gain is in the estimation of expected electricity under the recorded production conditions; it says nothing about plant energy use having fallen or about savings that could be achieved.

Giving the models a history of completed operations did not help. Under the tested representation of 16 same-line operations within the week, neither the history-enabled Ridge control nor TFT, N-HiTS or the tested TimesFM 2.5 and Chronos-2 configurations beat the current-input control on identical records, and the two pretrained pipelines showed instabilities on irregular industrial contexts (an extrapolating covariate adapter in one case, a device-dependent normalization in the other) that had more to do with covariate handling than with the pretrained weights. Production adjustment worked most consistently on Pellet 2 and Pellet 3; Pellet 1 improved on SEC but remained systematically underestimated and needs line-specific recalibration or additional predictors before any prospective use.

What the evidence supports today is retrospective assessment at the closure of each recorded operation: an expected value, an empirically calibrated residual interval and a traceable deviation that marks candidates for technical review. Prospective supervisory use will require validation on a new consecutive production period with prespecified line-level criteria, together with verified record identity, measurement boundaries, predictor availability and calibration rules. Until then, automatic control, modification of process set-points and operational digital-twin integration remain outside the demonstrated scope.

CRediT authorship contribution statement

Vladimir Sousa Santos: Conceptualization, Methodology, Investigation, Writing – original draft, Project administration, Funding acquisition. **Juan José Cabello Eras:** Conceptualization, Supervision, Resources, Writing – review & editing. **Eder Elías Herrera Rodiño:** Investigation, Data curation, Resources, Writing – review & editing. **Francisco Manuel Arrabal-Campos:** Methodology, Software, Formal analysis, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article (the full source-variable dictionary, descriptive statistics, the complete coefficient table of the frozen Ridge model, bootstrap and calibration details, residual-interval diagnostics by line and week, stratified and per-seed results of the temporal benchmark, CPU–GPU computational diagnostics and four additional figures) is provided as a separate document.

Data availability

The code, frozen model configurations, archived aggregate results, figure scripts and LaTeX sources are openly available in the GitHub repository https://github.com/fmarrabal/pelleting-energy-baseline-real-data (release `v1.0`). The industrial production and electricity records, and the record-level tables derived from them, belong to the plant and are not publicly available; they can be obtained

from the corresponding author (fmarrabal@ual.es) on reasonable request, subject to the data owner's agreement. Pretrained checkpoints are identified by repository and revision and are not redistributed.

Appendix A. Computational and numerical reproducibility

The GPU experiment used Python 3.13.13, PyTorch 2.14.0 with CUDA 13.0, NeuralForecast 3.2.2, the TimesFM package 3.0.2, Chronos-forecasting 2.3.2 and scikit-learn 1.9.0. The pretrained checkpoint revisions were `1d952420fba87f3c6dee4f240de0f1a0fbc790e3` for TimesFM 2.5 and `29ec3766d36d6f73f0696f85560a422f50e8498c` for Chronos-2. TimesFM received 16 observed context records, padded to a maximum context of 32, with a configured maximum horizon of 32; only its first forecast position was evaluated. Chronos used float32, disabled cross-query learning and returned the median forecast. Model configurations, checkpoints and record-level outputs are retained in the repository; pretrained weights are identified by revision rather than redistributed.

All 128,896 logged donor relationships passed the checks for same-pelleting-line membership, partition membership and completion before the target origin; the total includes repeated relationships across nested and expanding blocks. Source-data, frozen-model and final-prediction fingerprints remained unchanged, and the shared CPU and GPU data, window and encoding functions were checked for consistency. These checks concern the recorded completion-time proxy, since actual publication delays are not available in the source.

Within the GPU run, the 24 saved neural checkpoints were reloaded and compared with their stored inference, together with four Chronos covariate batch-size checks. All 28 checks passed the recorded tolerances with a maximum difference of 7.63×10^{-5} kWh. Fresh CPU and GPU neural fits were less closely aligned, as optimization can follow different trajectories even with the same integer seed: TFT's CPU and GPU MAEs were 53.25 and 53.65 kWh, and neither established an advantage over the matched Ridge control under the paired intervals. Figure A.10 shows all three GPU seeds and their ensemble.

For frozen Chronos-2 with covariates, the median absolute CPU–GPU prediction difference was 0.86 kWh and the maximum 57.04 kWh, and its overall MAE moved from 85.65 to 85.53 kWh. Maximum prediction differences for both frozen TimesFM variants and for univariate Chronos stayed below 1.9×10^{-4} kWh. To trace the covariate sensitivity, the installed Chronos preprocessing and instance normalization were applied to the same 584 contexts, giving 6,424 target/covariate

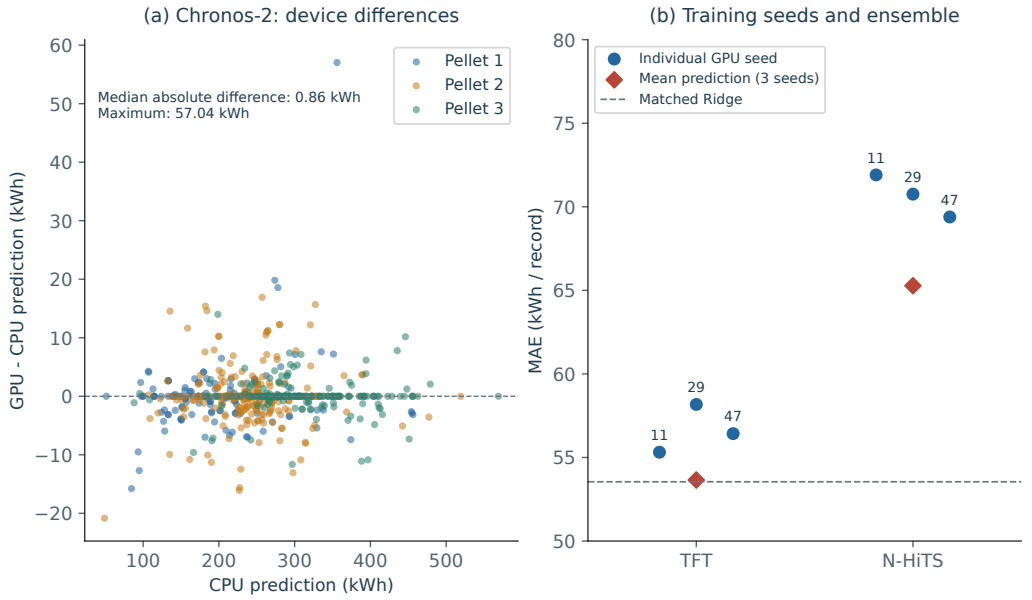


Figure A.10: Numerical and training variability. (a) Every paired CPU/GPU Chronos-2 covariate output as a signed difference against the CPU prediction. (b) All three GPU seeds (11, 29 and 47) and their prediction-mean ensemble for TFT and N-HiTS, with the matched Ridge reference; averaging predictions is not the same as averaging the three MAEs. These diagnostics accompany the primary comparison without selecting a preferred seed or device.

channels. Among the 1,097 constant historical channels, 694 showed a CPU–GPU difference above 0.1 in normalized context values, whereas no non-constant channel exceeded that threshold; the largest difference was 0.88137 in the transformed coordinates.

The implementation substitutes a small scale only when the calculated scale is exactly zero, so tiny round-off variances in constant channels can be amplified before the inverse-hyperbolic-sine transform. This mechanism is consistent with the observed sensitivity, although it is not an end-to-end causal demonstration for each prediction. The reported benchmark keeps the native preprocessing and original outputs; a corrective normalization policy would require a separately selected and evaluated experiment.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Trinka as a language-editing assistant to improve the readability and English of the manuscript, and

Claude Code as a coding assistant for writing and documenting the analysis scripts. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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